

Day 46



Local and Global Scope in JavaScript:

What is Scope?

Scope means where a variable can be accessed in your program. It decides which variables you can use and where.

Global Scope: A global variable is a variable that can be accessed from anywhere in the program.

How created?

- Declared outside any function/block
- Declared without let/var/const (not recommended)

Example: global variable

```
JS main.js > ...
1  let x = 10; // global variable
2  function show() {
3      console.log(x); // can access
4  }
5  show();
6  console.log(x); // accessible here too
```

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```
● PS E:\JavaScript\Day41-50\Day46> node main.js
10
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```

Local Scope: A local variable is only accessible inside the block/function where it is created.

Types:

- **Function scope** → created inside a function
- **Block scope** → created inside { } using let or const

Example: function local scope

```
8   function run() {
9       let a = 5; // local variable
10      console.log(a);
11  }
12
13  run();
14  // console.log(a); // ERROR → not accessible outside
```

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```
● PS E:\JavaScript\Day41-50\Day46> node main.js
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```

Example: block scope

```
15
16  {
17      let b = 20;
18      console.log(b); // OK
19  }
20
21  console.log(b); // ERROR
```

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```
● PS E:\JavaScript\Day41-50\Day46> node main.js
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E:\JavaScript\Day41-50\Day46\main.js:21
console.log(b); // ERROR
      ^
```

let, const, var Scope Difference (Simple)

Keyword	Scope Type	Important Note
var	function scoped	ignores block scope
let	block scoped	preferred
const	block scoped	cannot be reassigned

Example: Nested Scopes (Scope Chain): Inner scopes can access outer scope, but outer scope cannot access inner scope.

```
23  let a = 10;
24
25  function outer() {
26      let b = 20;
27      function inner() {
28          console.log(a); // OK
29          console.log(b); // OK
30      }
31      inner();
32  }
33  outer();
34  // b not accessible here → ERROR
```

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